LLM ENGINEERING

Week 01

Getting Started

Bethgelab · Thursday, 15 October 2026

University of Tübingen · Winter term 2026/27

Tübingen AI Center



> I believe that every university that has an AI group should train their own large language model.

Raia Hadsell

VP of Research, Google DeepMind

Frontier AI Innovation Summit, Stuttgart, 2026 (Hadsell, 2026)

Academic compute is scarce

- An envied lab server: 4 GPUs, €10,000
- A typical academic can use 4 GPUs for days (Khandelwal et al., 2025)
- Llama 3 405B trained on up to 16K H100s (Grattafiori et al., 2024)
- Using compute well sets you apart



Frontier-lab interviews test engineering

- Interviews at OpenAI, Anthropic, Google DeepMind, Meta Superintelligence Labs
- Questions mostly on parallelization and hardware utilization
- Weeks 2–6 teach both



The goal:

the skills to start a PhD in a top program,
join a frontier lab such as Google DeepMind,
or found a deep-tech startup,
and be productive from day 1.



What does it take to be productive from day 1?

- Where the field stands
- Derisk every decision before scaling
- How this course trains you
- The one objective: next-token prediction

Every week runs one loop

Specification	Implementation	Tests + measurement	Tutorial feedback

- One config-driven codebase grows all semester
- Coding starts in Week 2

Seven eras since 2017

- The Transformer, the core architecture, dates from 2017 (Vaswani et al., 2017)
- Many milestones map to a course week

Should you train your own model?

When would you, when not,
and what would you check first?



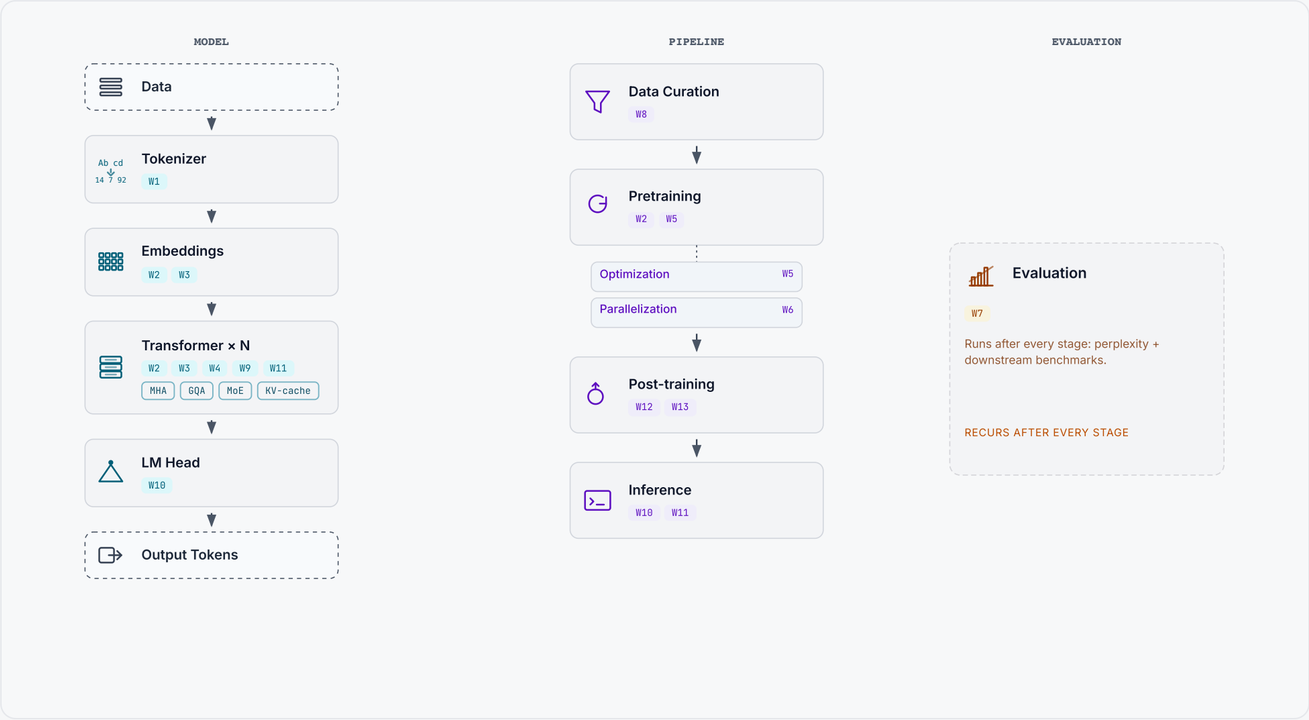
Train your own model only with a reason

- Research: answer a falsifiable question
- Production: constraints no existing model meets
- Openness: fill a specific unmet need
- Learning: build it to understand it
- First: benchmark open models under your constraints



One change reaches across the pipeline

- The tokenizer sets sequence length and embedding table size
- Attention sets memory, throughput and the state inference keeps
- Data sets the distribution the model learns
- Evaluation choices change which conclusions look favorable



10 CONNECTIONS

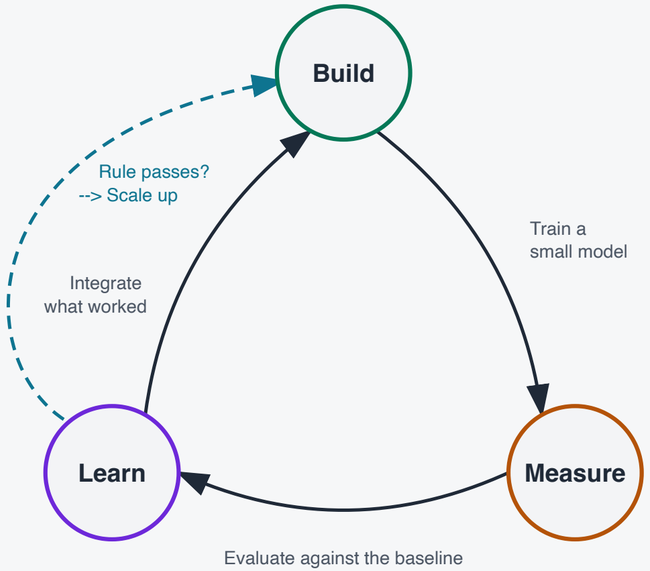
◀ Prev | Reset | Start ▶

Click "Start" to walk through how decisions in one part of the pipeline shape every other part.

Declare the experiment before you run it

- Reasoning alone cannot tell whether a change helps (Ben Allal et al., 2025)

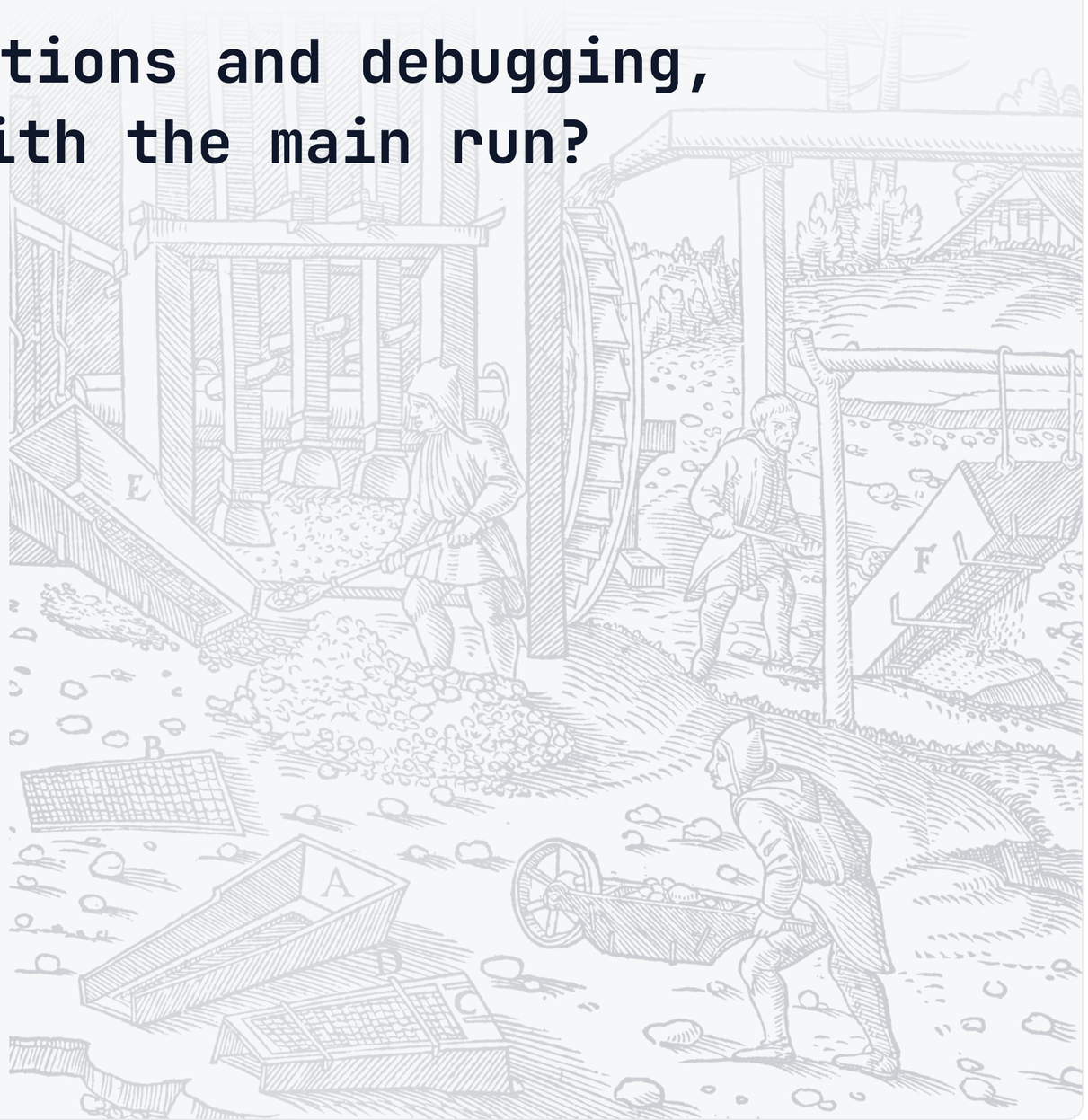
Field	Declare before the run
Hypothesis	intervention and the metric change you expect
Control	trusted baseline and one intended change
Budget	matched tokens and compute
Evidence	fixed evaluation set and validated metric
Noise	seeds or another variance check when needed
Decision	threshold, set before the run, to scale up, revise or stop



How much did derisking cost?

SmolLM3's ablations and debugging, compared with the main run?

- 1: 5%
- 2: 10%
- 3: 25%
- 4: over half



Derisking cost more than half the main run

- 1: 5%
- 2: 10%
- 3: 25%
- 4: over half

Ablations and debugging	Main run
161,280 GPU-hours	276,480 GPU-hours

- 58% of the main run, 37% of the two combined (Ben Allal et al., 2025)
- Small runs must test the same decision

You learn by building one codebase

- Learn by implementing
- One codebase grows across the semester
- Small experiments, real engineering practice
- Revisit ideas: recall them and compare them with new ones
- Setup is provided, so your effort goes to mechanisms

From one block to post-training in 14 weeks

Phase	Weeks	Outcome
Orientation	1	LM objective, tokens, environment
Foundation	2-4	Decoder block, multi-head attention + rotary position embedding (RoPE), modern attention
Training at Scale	5-7, 9	Optimization, parallelization, scaling, data
Advanced Systems	8, 10, 13	Architecture experiments, inference, mixture of experts (MoE)
Post-Training	11-12	Supervised fine-tuning (SFT), direct preference optimization (DPO), and on-policy RL
Capstone	14	Synthesis, last project pitches

- Coding exercises in Weeks 2–13, all in one codebase

The reading is mandatory

- Read on your own: reading slides, notes, bonus chapters
- Quizzes test the full script
- Review the Anki deck every week
- Quiz questions come from the Anki cards
- Lectures: you predict and discuss before results appear

Assessment

Component	Weight	Details
Final project	80%	Reproduce, investigate or contribute
Bi-weekly quizzes	20%	Best 5 of 6

- 9 ECTS
- Pass 6 of 12 exercises to qualify for the project
- Quizzes: Weeks 3, 5, 7, 9, 11, 13
- Each quiz: about 10 minutes, in the tutorial, cumulative
- Exercises due Wednesday 22:00, before the lecture

The final project

- Reproduce, investigate or contribute
- Groups of two
- Pitch in a tutorial, by Week 14
- Code, report and talk are due after the semester
- Details on how projects work follow by Week 5
- Start collecting ideas now

About me

- PhD student, Bethgelab, Tübingen AI Center
- Scalable oversight of language models
- Post-training with dense intrinsic rewards
- Studied in Münster and Tübingen



Dates, rooms and links

	When	Where	Who
Lecture	Thu 14:15–16:00	Lecture hall A2	Me
Tutorial I	Thu 12:15–13:45	A-301, 3rd floor	Lab PhD students, rotating
Tutorial II	Fri 14:15–15:45	A-301, 3rd floor	Me

- All rooms: Maria-von-Linden-Str. 1
- Lecture site: llm.strueber.dev
- Exercises: github.com/LLM-Engineering-26-27
- Announcements: ILIAS course
- Discord server: invite link on ILIAS

Pairs, accounts and the cluster

- 32 places (16 pairs), over 80 signups
- Random draw tomorrow among eligible candidates
- Drawn? We add you to the ILIAS course
- Partner deadline: after next week's tutorial
- A GitHub account for your pair repository
- Cluster account: step-by-step PDF on ILIAS

AI assistants

- Write the graded code yourselves
- Use assistants for explanations, infrastructure and plots
- Final project: allowed if you disclose how
- Quizzes: no tools
- You must explain every line

The first tutorial is next week

- Questions on Week 1 and Bonus 1
- Help with the cluster sign-up
- Everyone leaves with a partner
- Week 2 exercise due Wed, 28 October, 22:00

> ⚡ **Due this week**
>
> Nothing is due this week



Text becomes token IDs

- A **tokenizer** cuts text into pieces and maps each to an ID
- Each ID selects one embedding row
- The course tokenizer has 8,192 vocabulary entries (PleIAs, 2025)
- Bonus 1 covers tokenization in depth

```
"The price is 342 euros."
→ ["The", "Ġprice", "Ġis", "Ġ3", "42", "Ġeuros", "."]
→ [307, 4614, 313, 356, 4738, 7057, 17]
→ 7 embedding rows
```

One objective to learn them all

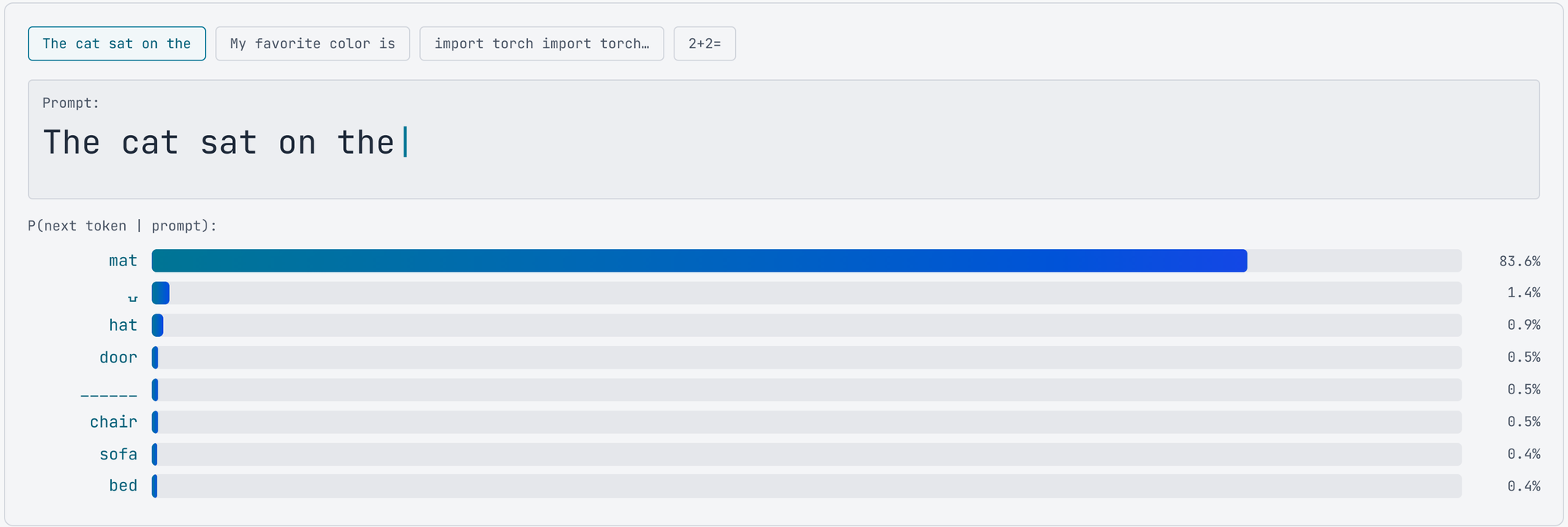
- `import torch.nn as → nn`
- `The capital of France is → Paris`
- `2+2= → 4`
- Assistants add post-training on top (Ouyang et al., 2022)

Which of these prompts has the most uncertain next token?

- 1: The cat sat on the
- 2: My favorite color is
- 3: `import torch.nn as`
- 4: `2+2=`

Uncertainty depends on the context

- 1: The cat sat on the
- 2: My favorite color is
- 3: `import torch.nn as`
- 4: `2+2=`



- Most uncertain: the favorite color
- A fixed phrase gets one confident answer

Logits become a distribution over the vocabulary

- One **logit** per vocabulary entry (V) at each position: `[B,S,V]`
- Softmax along the vocabulary axis
- Training rewards the probability of the actual next token

(1)

$$p_\theta(x_{i+1} = v \mid x_{\leq i}) = \mathrm{softmax}(z_i)_v = \frac{e^{z_{i,v}}}{\sum_{u=1}^{V} e^{z_{i,u}}}$$

The chain rule factors every sequence

$$P(x_1, x_2, \ldots, x_n) = \prod_{t=1}^{n} P(x_t \mid x_{<t}) \tag{2}$$

- Autoregressive factorization: an exact identity, by the chain rule
- By convention, the first token conditions on a beginning-of-sequence token
- A prediction sees only its prefix

Training minimizes cross-entropy

(3)

$$\mathcal{L} = -\frac{1}{N} \sum_{(b,i) \in \mathcal{M}} \log p_\theta(y_{b,i} \mid x_{b,\le i})$$

- The set $\mathcal{M}$ holds the non-padding targets, and N counts them
- Probability 0.8 on the target costs $-\log 0.8 \approx 0.223$
- Minimizing it maximizes log-likelihood

```
inputs = raw[:-1]
labels = raw[1:]
```

What questions does this raise?

A base model learns from this one objective and nothing else.

$$\mathcal{L} = -\frac{1}{N} \sum_{(b,i)\in\mathcal{M}} \log p_\theta(y_{b,i} \mid x_{b,\leq i}) \tag{3}$$

Most of your questions have a week

- Which data to learn from: Week 9
- Which model predicts well: Weeks 2–5, 13
- Using the hardware well: Weeks 4–6
- Measuring a model: Week 7
- Shaping behavior: Weeks 11–12
- No week? A final-project idea

What it takes to be productive from day 1

- Specify, build, measure, get feedback
- The field has moved fast through seven eras since 2017
- Derisk with one change, a matched budget, a declared threshold
- One codebase, 14 weeks, nothing due now
- Text becomes token IDs, which select embedding rows
- The chain rule leads to the next-token cross-entropy loss



To-dos until next week

Muddiest point

Do it right now! (2 min)

Intro survey

Do it at home.

Once you have a spot in the class

- Pick a tutorial slot on ILIAS
- Sign up for the cluster: it takes a few days
- How-to for the cluster follows tomorrow

Before the tutorial

- Read Bonus 1: tokenization
- Think about the reflection questions of Week 1 and Bonus 1
- Write down one question to discuss

Reflection questions

1. A team can explain its training system on paper, but its first multi-GPU run is unstable. How would you use the course's specification–implementation–measurement–feedback loop to decide what to inspect or change next? What evidence would keep the process from becoming trial and error?
2. A hospital wants a model that drafts discharge letters in German, and patient records may not leave its servers. Before anyone trains a model, write the decision record you would give the hospital: how you would decide what quality is good enough, how you would find out what the open models the hospital can run on site achieve, how you would tell a real difference from noise on a test set of a few hundred letters, and which result would make you recommend training your own model, adapting an existing one, or stopping.
3. A data-mixture change improves loss in a small, short-context ablation. What must remain comparable before treating that result as evidence for a larger, longer-context run, and what finding would make you withhold the scale-up decision?
4. Two tokenizers represent the same corpus with different average sequence lengths and vocabulary sizes. Trace two downstream consequences and design an evaluation that could justify choosing one for a constrained deployment.
5. A model's pretraining loss improves smoothly, but a benchmark gain appears under one prompt and threshold and disappears under another. What can the team conclude about prediction and capability, and what remains uncertain?